Battery technology solutions from Mine to Lab to Line to Landfill !

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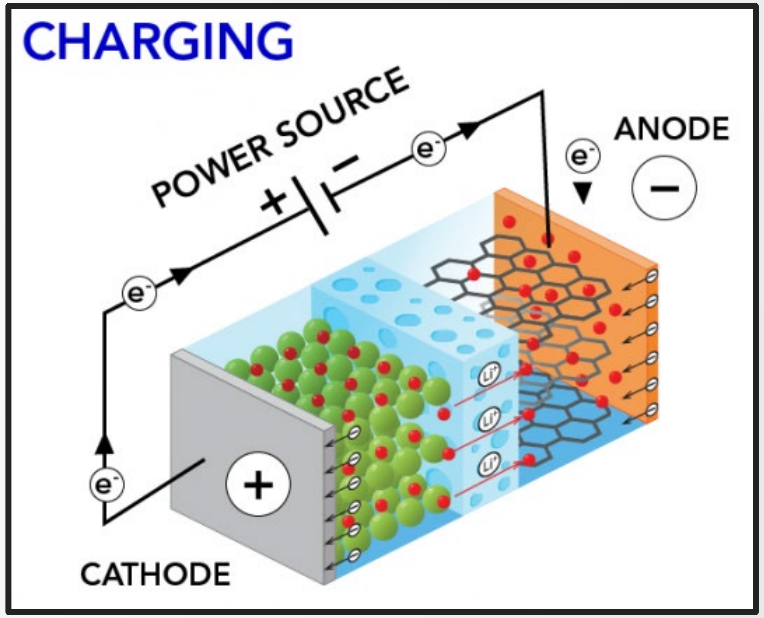
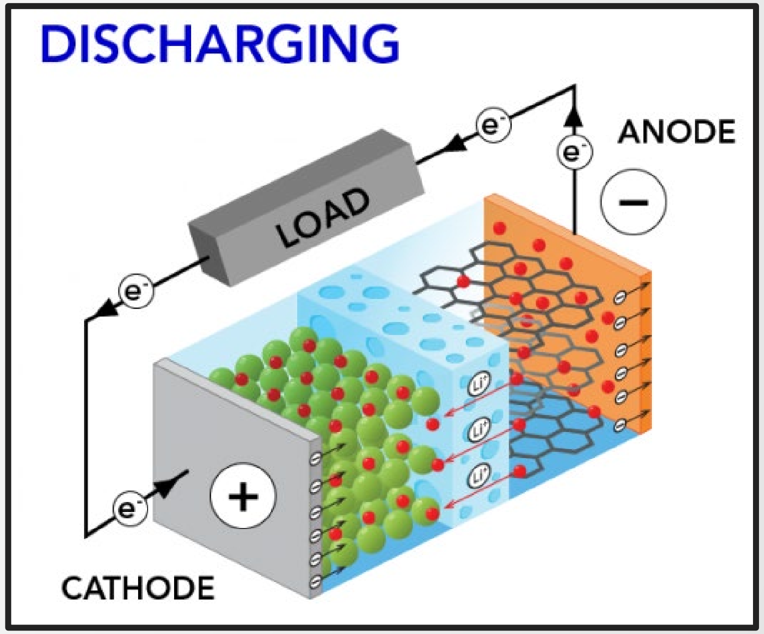
Head : Clean Energy

The world leader in serving science

Overview of the presentation

- Development of the battery market
- Rise of the Gigafactories
- What makes up the battery supply chain?
- Why is analysis needed in battery manufacturing?
- Analysis solutions across the supply chain
- Battery technology resources from Thermo Fisher Scientific
- Thermo Fisher Scientific at the exhibition



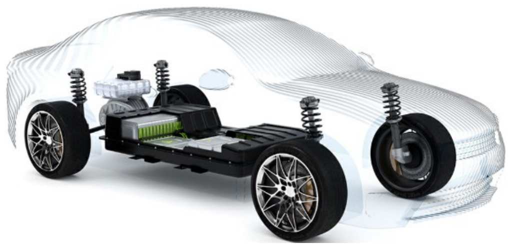


Development of the battery market

Rapid increase in battery production mainly due to electric vehicles (EV's) and static energy storage systems

Market estimates indicate >15-fold increase in EV sales from 2020 to 2030*

Estimated required global annual battery production (in GWh) increase of >25-fold between 2020 and 2030**



EV in 2020	EV in 2025	EV in 2030
3 million	20 million	47 million

GWh in 2020	GWh in 2025	GWh in 2030
~300 GWh	~2500 GWh	~8000 GWh

* International Energy Agency web site - https://www.iea.org/data-and-statistics/charts/global-ev-sales-by-scenario-2020-2030

** Benchmark Mineral Intelligence Battery Gigafactories database

Rise of the Gigafactories

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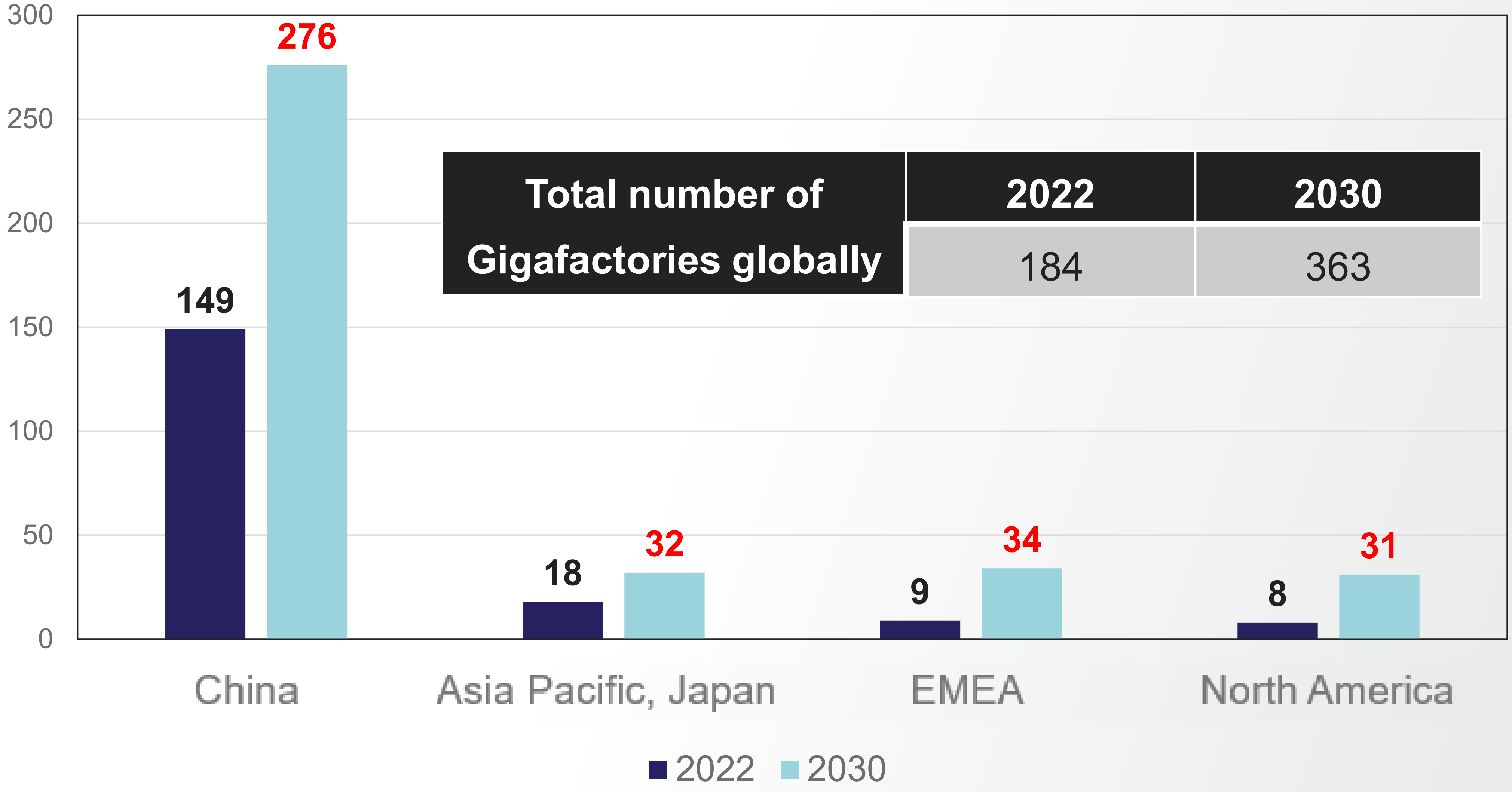

Total number of Gigafactories globally	2022	2030
	184	363

Source: Benchmark Mineral Intelligence Battery Gigafactories database

What makes up the battery supply chain? - 1

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- ***Research***
  - Material characterization, studying causes of battery failure, investigating new battery chemistries and technologies
- ***Raw material production***
  - Mining and refining of ore, mineral and brine feedstocks
- ***Cell material manufacture***
  - Cathode active material (CAM), electrolyte, anode, binder, separator materials
- ***Battery manufacturing***
  - Electrode calendering, cell production

What makes up the battery supply chain? - 2

- ***Battery performance testing***
  - QA/QC on finished cells, post-mortem failure studies

- ***Recycling***
  - Screening incoming material; composition and purity certification of end products

- ***Environmental and health and safety analysis***
  - Meeting local environmental emission and occupational health regulations

- ***Data handling and reporting***
  - Streamlining data handling from sample receipt to analysis report using Laboratory Information Measurement Systems (LIMS)

Serving the entire battery value chain

Capabilities from lab-to-line

	RESEARCH	MINERAL PROCESSING	RAW MATERIALS CONTROL	BATTERY PRODUCTION	BATTERY QUALITY ASSURANCE	RECYCLING
	Applied research functionalized materials	Bulk production of lithium, copper, cobalt, nickel, graphite	Ensuring raw materials meet manufacturing specifications	Manufacture of cells and cell arrays	Ensuring finished product meet specifications	Reuse, recycling and recovery of hazardous raw materials
Analytical need	**Characterization and failure analysis**	**Bulk materials refinement**	**Quality by design and material rejection**	**Process control & in-line inspection**	**Failure analysis and rejection**	**Inspection and characterization**
Key technologies	• Spectroscopy • Microscopy • Mass Spectrometry • Trace Elemental Analyzers • Rheology • Extruders • Equipment Cooling Analysis • Chemicals & Lab Supplies & PPE • Imaging Data Visualization and Analysis • Materials and Properties Characterization • LIMS	• Mineral & Ore Processing Elemental Analyzers • Trace & Bulk Elemental Analyzers • Hand-held Spectroscopy • Spectroscopy • Microscopy • Chromatography • PPE • Imaging Data Visualization and Analysis • Materials and Properties Characterization • LIMS	• Hand-held Spectroscopy • Materials Characterization • Scanning Electron Microscopy • Trace & Bulk Elemental Analyzers • Viscometers • Rheology • Weighing & Monitoring • PPE • Imaging Data Visualization and Analysis • Materials and Properties Characterization • Reporting • LIMS	• In-line Thickness & Basis Weight Measurement • In-line X-ray inspection & NDT • Microscopy • Electrode Drying • Hand-held XRF • Environmental Control • Equipment Cooling • Water Purification • PPE • Imaging Data Visualization and Analysis • Materials and Properties Characterization • LIMS	• Electrochemistry Systems • Microtomography • Microscopy • X-ray inspection & NDT • Temperature Control • Trace & Bulk Elemental Analyzers • PPE • Defect analysis and reporting • Imaging Data Visualization and Analysis • Materials and Properties Characterization • LIMS	• Hand-held Spectroscopy • Microscopy • Spectroscopy • Trace Elemental Analyzers • X-ray inspection & NDT • PPE • LIMS

Why is analysis needed in the battery supply chain?

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- Safety
  - Minimising fire risk from short circuits or uncontrolled discharge
  - Protecting workforce in manufacturing environments
- Ensuring cell performance
  - Certifying quality of materials used in cell production
- Improving battery technology
  - Reducing defects, advancing battery design, investigating and developing new battery materials
  - Maximising charge carrying capacity, lifetime and charging speed
- Decreasing costs
  - Accelerating production, improving manufacturing consistency and decreasing scrap rate

Analysis solutions across the supply chain

Mining and refining

- Conveyer belt ore weighing systems
- High throughput system optimised for:
  - Production rate measurement, metallurgical accounting and precision blending
- Ore sorting to differentiate material that is ≤ cut-off grade
  - Mine grade control, material sorting, material blending and stockpile control
- Element and anion compostion and impurity analysis
  - Lithium minerals, ores and brine
  - Refined lithium hydroxide and lithium carbonate
- Accurate, low detection limit analysis required
- Robust and reproducible operation essential

Thermo Scientific™ CB Omni™ Agile Online Elemental Analyzer

Thermo Scientific™ Ramsey™ Series 17 Belt Scale System

Thermo Scientific™ iCAP™ PRO ICP-OES

Thermo Scientific™ iCAP™ RQplus ICP-MS

Thermo Scientific™ Dionex™ ICS-6000 HPIC™ System

Compounding, extrusion and rheology measurements

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- Production of anode, cathode and separator films
- Twin-screw extruders compound PTFE / PVDF binder and cathode / anode materials to produce solvent-free slurries
- High shear ability enables formation of thin PTFE fibres that effectively bind together cathode / anode particles
- Separator films produced in a single process without segregation or coalescence
- Slurry viscosity over shear rate flow curves provide insight of flow behavior during slurry pumping, stirring, and coating
- Measurement of slurry viscosity and shear rate necessary for ensuring efficient coating of slurries onto electrode substrates and for developing new formulations
    - Optimize charge storage capacity and lifetime

Thermo Scientific™ Process 11 Twin Screw Extruder


Thermo Scientific™ Process 16 Twin Screw Extruder


Thermo Scientific™ HAAKE™ MARS™ 40 / 60 Rheometer


Thermo Scientific™ HAAKE™ MARS™ iQ Rheometer

Cathode, electrolyte and anode material analysis

- Metal and semi-metal element and anion compostion and impurity analysis
  - NMC, LCO, NCA and LFP cathode active materials
  - Graphite, silicon and composite anode materials
  - Novel anodes such as Nb doped materials
- C, H, O, N and S measurement in battery materials
- Organic solvent composition and impurity analysis in electrolyte materials
  - Organic carbonates
  - Tetrafluoroborate, perchlorate, and hexafluorophosphate anions, plus associated impurities
- Provides the accuracy, precision, sensitivity and robustness required for battery material QA/QC

Thermo Scientific iCAP PRO ICP-OES

Thermo Scientific iCAP RQplus ICP-MS

Thermo Scientific Dionex ICS-6000 HPIC System

Thermo Scientific™ TRACE™ 1610 GC

Thermo Scientific™ ISQ™ 7610 GC-MS

Thermo Scientific™ Flash*Smart*™ OEA

Bulk elemental analysis in solid battery materials

- Quantifying elements in solid materials and thin films
- Surface analysis of materials such as cathode alloys
- Accelerates analysis as avoids sample digestion
  - Fast sample turnaround for production environments
- Ideal for samples that are difficult to decompose
  - Graphite, silicon and novel anode materials
- Rapid screening capability with handheld systems
- Capability for crystallinity, stability, reactivity, texture and stress analysis of battery materials
- Fast and accurate results, enabling production issues to be identified and quickly corrected

Thermo Scientific™ ELEMENT™ GD Plus GD MS

Thermo Scientific™ ARL™ QUANT'X EDXRF Spectrometer

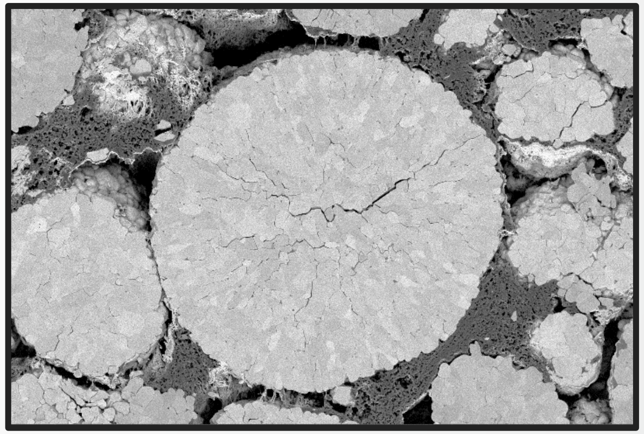
Thermo Scientific™ ARL™ EQUINOX 100 XRD

Thermo Scientific™ ARL™ 9900 XRF

Thermo Scientific™ Niton™ XL3t GOLDD+ XRF

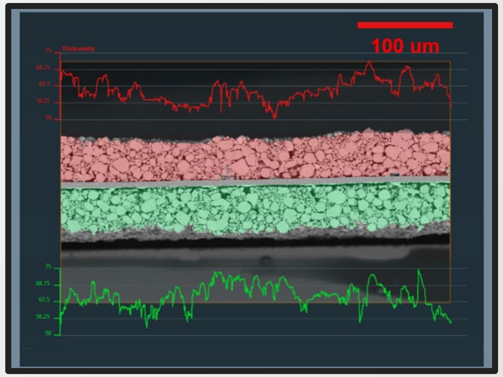
Material characterisation – electron microscopy (1)

- Rapid, high resolution 2D imaging and analysis
  - Structural characterization of different battery components
- Quality control of cathode and anode particles
- Nanometer resolution, optimized for battery samples
- Inert Gas Sample Transfer (IGST) capability for both scanning and transmission electron microscopes (SEM / TEM)
  - Enables analysis of air-sensitive materials in their native state
- High quality observation / characterization of materials requires artifact-free surface, achieved using broad ion beam polishing



Thermo Scientific™ Axia™ ChemiSEM™ SEM-EDS and CleanMill™ Broad Ion Beam System

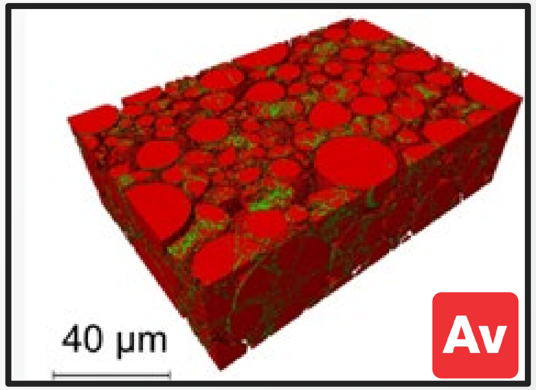
CleanConnect™ IGST System



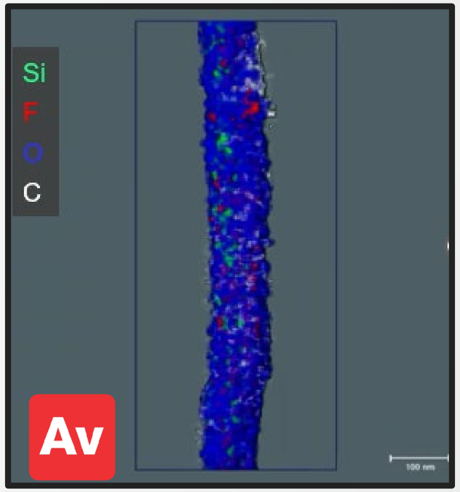
Battery material imaging with Avizo Software

Material characterisation – electron microscopy (2)

- Dual beam functionality for battery materials analysis
  - In-situ polishing and imaging of battery materials in the same system
- 2D and 3D imaging at the cell, electrode and particle level
  - Atomic scale characterization via TEM
  - Nanometer resolution with high contrast in 3D using DualBeam operation
- Structural characterization of the solid electrolyte interphase (SEI) layer and different active material chemistries
- Direct imaging of Li atoms in TEM using iDPC capability
- Visualization, image segmentation, and modeling based on 2D and 3D datasets using in house software solutions



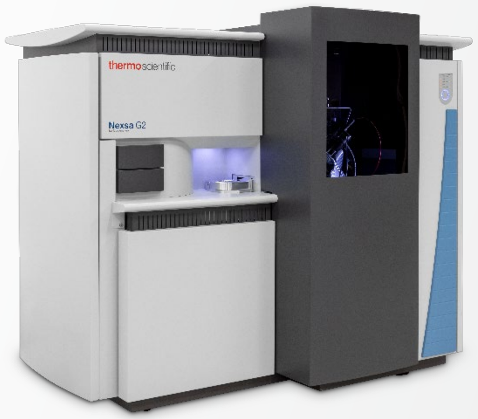
Thermo Scientific™ Helios™ 5 PFIB DualBeam™ SEM



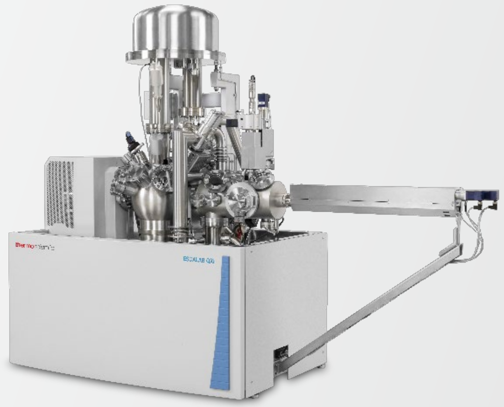
Thermo Scientific™ Talos™ F200X G2 TEM

Material characterisation – X-ray photoelectron spectroscopy (XPS)

- Surface chemistry analysis technique
- Elemental and chemical state information about the top layer(s) of a material
  - Images the top ~10 nm of sample
  - Quantified chemical state analysis
  - Depth profiling of hard and soft materials
  - Vacuum transfer module for air-sensitive materials
  - Enables changes in cathode / anode material composition following charge cycling to be detected
  - Essential technique for understanding interfaces between electrolytes and electrodes

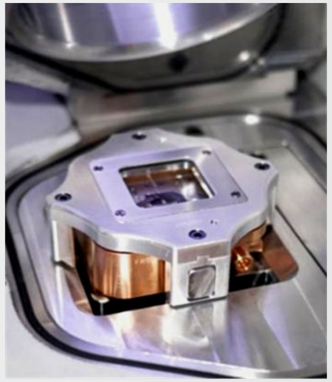


Thermo Scientific™ Nexsa™ G2 XPS



Thermo Scientific™ ESCALAB™ QXi XPS

Thermo Scientific™ K-Alpha™ XPS



Vacuum transfer module

Material characterisation – Infra-red and Raman spectroscopy

- Spectral fingerprinting for rapid identification of unknowns in solids, liquids and gases
- In operando and in situ sampling, real time analysis capabilities
  - Analysis of cathode / anode slurries with probe or in a container
- Cathode / anode - organic electrolyte interface studies
- Moisture content analysis in solvents and lithium compounds
- Compact system available for use in gloveboxes for air sensitive sample analysis
- Visualize chemical compositional distribution on component surfaces and profile chemical changes during battery cycling
- Evaluate spatial distribution of phases in a sample
- Ideal for studying allotropes of carbon and transition metals

Thermo Scientific™
Nicolet™ Summit™
FTIR Spectrometer

Thermo Scientific™
Nicolet™ iS50
FTIR Spectrometer

Thermo Scientific™
DXR3 Raman
Microscope

Thermo Scientific™
Antaris™ II
FT-NIR Analyzer

Other material characterization and failure analysis solutions

- Ultra-trace analysis of elemental impurities and metal-based degradation products
  - Sub-ng/g impurity detection in multi-component battery materials
  - Hybrid technologies for identifying metal based organic species formed during battery cycling
  - Increased understanding of interactions between electrode materials and electrolyte
- Complex electrolyte degradation pathway elucidation
  - Identification of species via accurate molecular mass determination
  - Flexible analysis using full range of hybrid technologies
  - Enables development of new battery chemistries to slow degradation and improve battery performance and lifetime

Thermo Scientific™ iCAP™ TQe ICP-MS

Thermo Scientific™ Dionex™ Integrion™ HPIC with Thermo Scientific™ ISQ™ EC/EM Mass Spectrometer

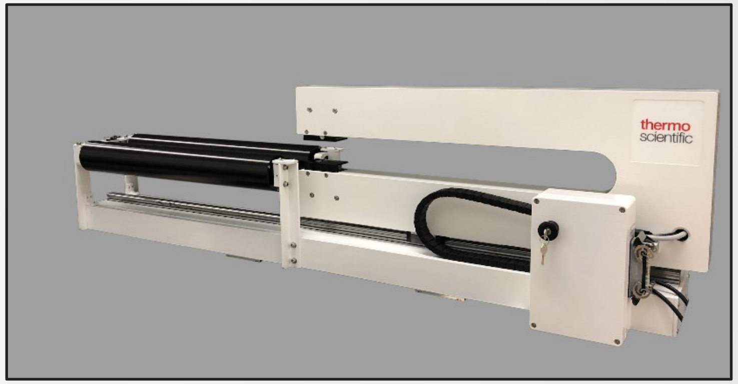
Thermo Scientific™ Orbitrap™ Exploris™ GC 240 MS

Thermo Scientific™ Vanquish™ UHPLC Systems and Thermo Scientific ISQ EC/EM Mass Spectrometer

Electrode calendering process inspection

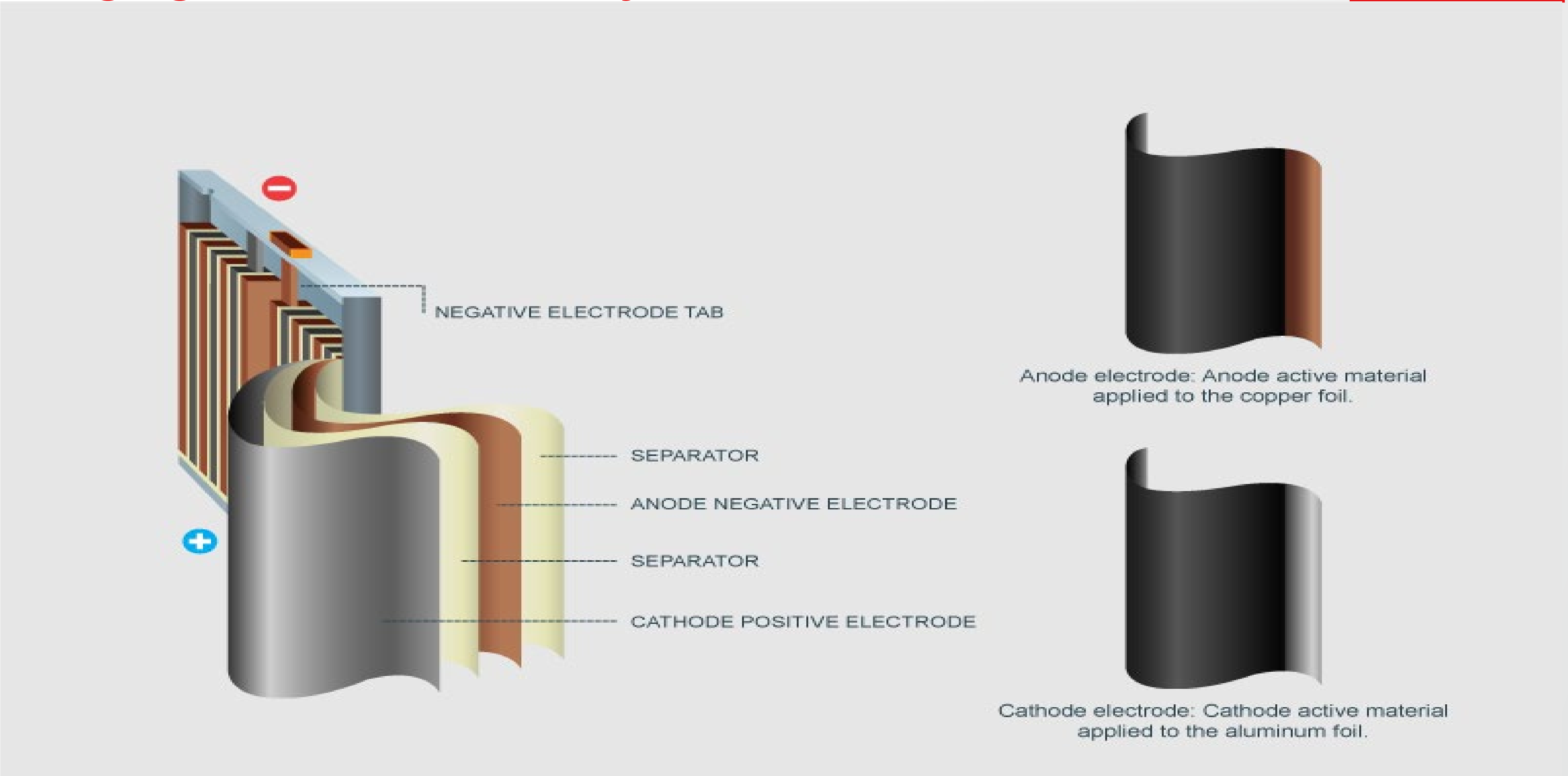
- Constant and repeatable electrode thickness is essential
  - Uneven coating on electrode substrates compromises charge density, recharge time, operational lifetime and safety
  - Defect detection vital to reduce waste and downtime, improve yields, ensure consistent product quality, and reduce failure in the field
- Calendering process with X-ray gauge monitoring ensures consistent compression of the coated electrode
  - Ensures correct electrode thickness, weight and particle size to control coating porosity, electrical contact and adhesiveness
  - Maintains accurate electrode dimensions
- Control of multi-layer separator film production
  - Separator film porosity, thickness and tensile strength influences battery self-discharge rate, lifetime and safety, so needs to be controlled
- Corrective actions require real-time data collection and visualization

Thermo Scientific™ LInspector™ Gauge with INTEGRA O-Frame Scanner for separator film measurement



Thermo Scientific LInspector Gauge with INTEGRA C-Frame Scanner for electrode calendering

Gauging solution for battery



Solving battery production challenges

Improving early detection to reduce scrap and improve productivity

Typical defects	
1	Non-uniform coating
2	Blisters / agglomerates
3	Pinholes / divots
4	Metal particle contaminants
5	Fractures / splits
6	Folds
7	Oxidation

Detection coverage

Typical in-line gauge covers less than 2% surface area of electrodes and separator film

Resolution & precision

Poor detection and identification of defects

Services and support

Impacts productivity and cost

Eliminate Defects - Reduce Cost

Recycling

- Vital to ensure batteries fully discharged before opening
  - Placed in discharge tank for defined period
  - Evolved gases and contaminant build up in the tank measured
- Black mass material extracted during first recycling stage requires screening to confirm battery chemistry
  - Sort incoming material to avoid extra processing steps
- Product composition and purity measured at each step in the recycling process
  - Elemental, anionic and organic components
- Sub-ng/g impurity levels required in final recycled products
  - Certificate of Analysis to confirm materials are battery grade

Thermo Scientific
iCAP PRO
ICP-OES

Thermo Scientific
iCAP RQplus
ICP-MS

Thermo Scientific
Dionex ICS-6000
HPIC System

Thermo Scientific
Trace 1610 GC

Thermo Scientific
ISQ 7610 GC-MS

Thermo Scientific
Niton XL3t GOLDD+
XRF

Environmental analysis and health and safety compliance

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- Monitoring, controlling and reporting emissions mandatory for battery cell and battery component material manufacturers
  - Element detection and quantification in water and soils
  - Organic and anionic compounds in waste streams and water
  - Regulated components in gaseous emissions
- Compliance with enviromental analysis regulations
  - EPA, ISO, DIN and the EU Water Framework Directive
  - WEEE and RoHS
- Compliance with occupational exposure regulations
  - REACH Restriction 71: Guideline for users of NMP
  - DIN EN 482: Workplace exposure - Procedures for the determination of the concentration of chemical agents – Basic performance requirements

Thermo Scientific iCAP PRO ICP-OES

Thermo Scientific iCAP RQplus ICP-MS

Thermo Scientific Dionex ICS-6000 HPIC System

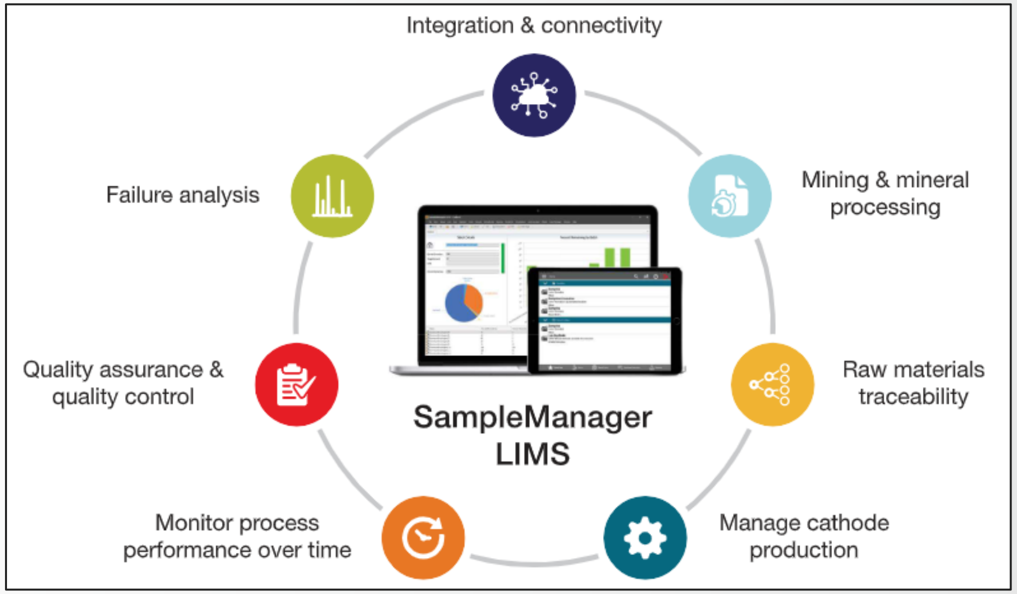
Thermo Scientific Trace 1610 GC

Thermo Scientific ISQ 7610 GC-MS

Thermo Scientific™ Prima PRO Process MS

Data handling and reporting

- Laboratory Information Management System (LIMS)
- Complete sample management and batch traceability from raw material to finished goods
- System and instrument integration to help optimize production technology operations
- Document, report and archive instrument calibration and maintenance records
    - ISO 17025 and ASTM compliance:
- Helps facilitate repeatable and reliable QA / QC testing workflow
- Enables maintenance of data integrity and compliance with lithium battery specific regulatory standards such as ISO/TC 333 - Lithium, and ISO 12405-2:2012



Laboratory consumables and other equipment: Fisher Scientific

ThermoFisher SCIENTIFIC

Lab essentials

Incubators, titrators, viscometers, microscopes, plastic and glassware, analytical balances, electrochemistry meters, carts and furniture, pumps, ovens, pipettes, pipette tips etc.

Chemicals and reagents

Cathode / anode materials, organic electrolyte solvents, inorganic salts ($LiOH$, Li_2CO_3, $LiPF_6$), separator / binder materials, slurry solvents (NMP) etc.

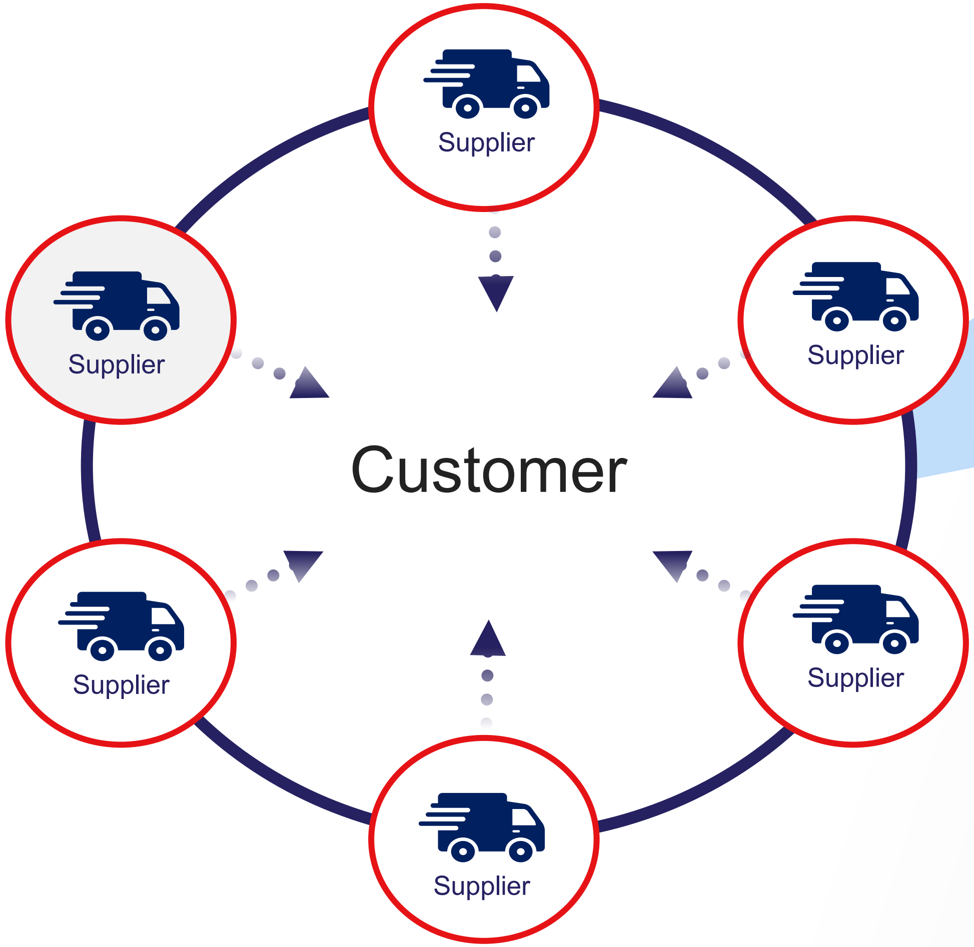
Safety

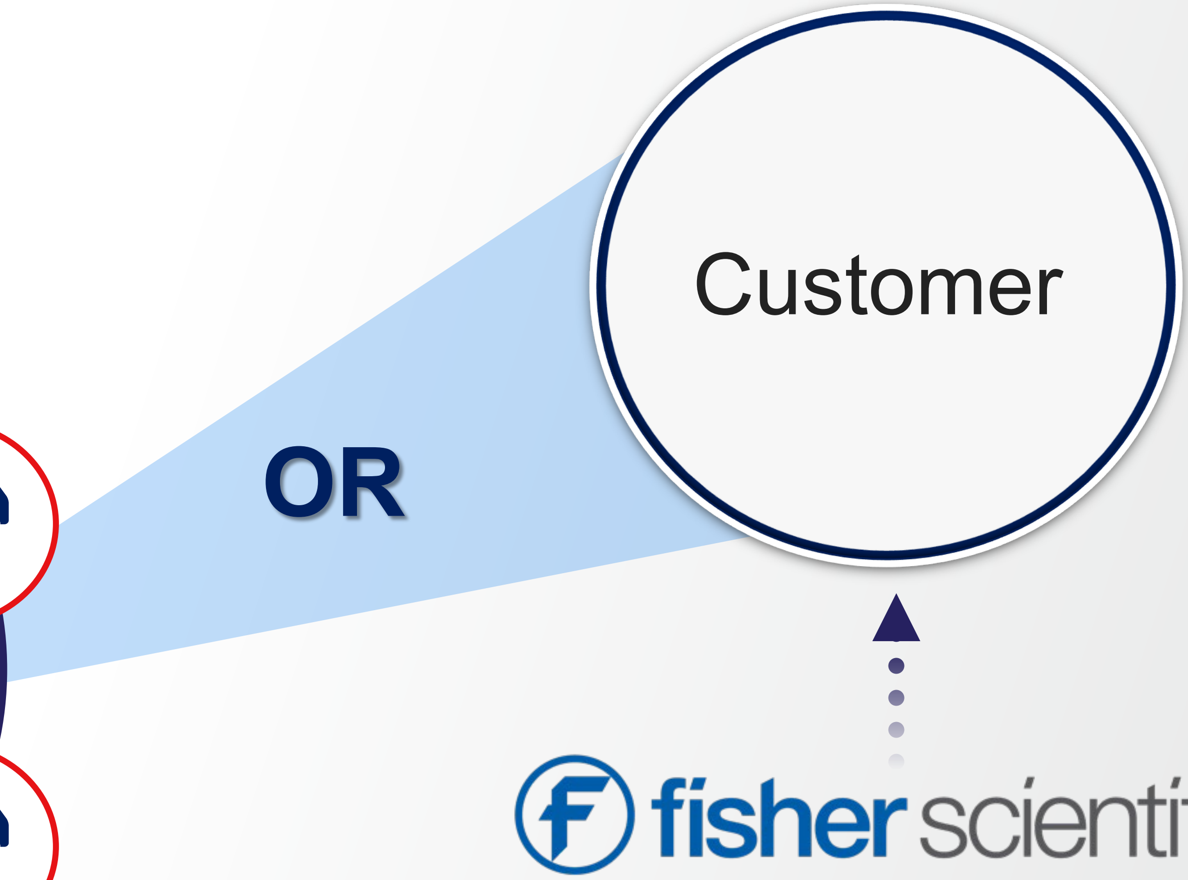
Lab coats, aprons, safety glasses, gloves and glove boxes, respiratory protection, cleaning supplies, wipes and pads, hazardous materials storage and disposal, cleanroom supplies

ONE supplier servicing ALL your product requirements = ONE point of contact and ONE process

Integrated value proposition

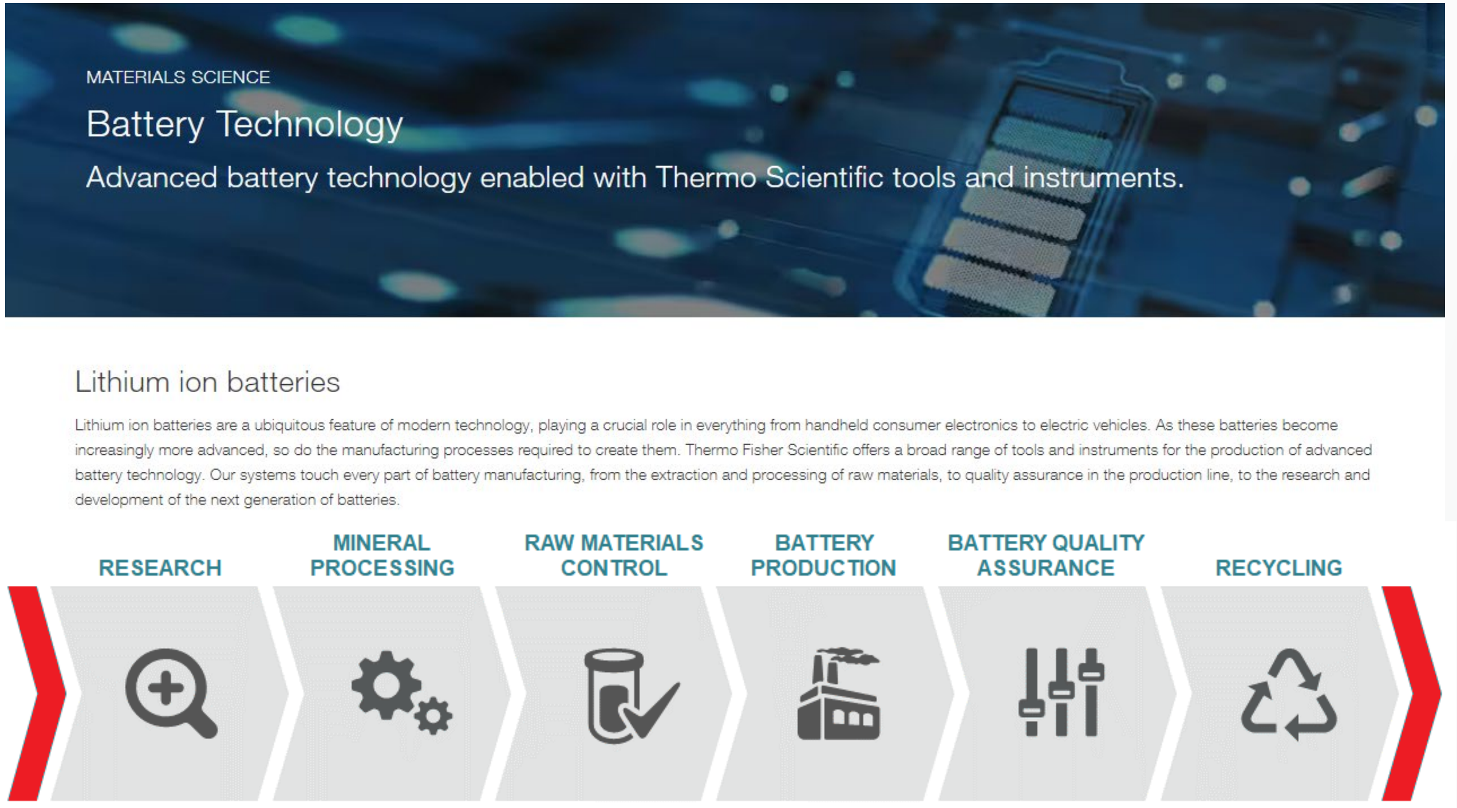
Multiple suppliers =
Multiple contacts and processes





ONE supplier servicing **ALL** your product requirements = **ONE** point of contact, and **ONE** process

Battery technology resources from Thermo Fisher Scientific



MATERIALS SCIENCE

Battery Technology

Advanced battery technology enabled with Thermo Scientific tools and instruments.

Lithium ion batteries

Lithium ion batteries are a ubiquitous feature of modern technology, playing a crucial role in everything from handheld consumer electronics to electric vehicles. As these batteries become increasingly more advanced, so do the manufacturing processes required to create them. Thermo Fisher Scientific offers a broad range of tools and instruments for the production of advanced battery technology. Our systems touch every part of battery manufacturing, from the extraction and processing of raw materials, to quality assurance in the production line, to the research and development of the next generation of batteries.

Thank you